Do System One Models Make Better Rerankers?

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Abstract. System One models, or decision models, answer a question about an input with a calibrated probability. They are only a month old but much discussed, and first reports have tried them as rerankers. For six Qwen foundation models (0.6B to 27B parameters) and ten test collections, we compare the untuned model with its decision fine-tunes and its reranker fine-tunes, one of which we trained ourselves. The reranker fine-tunes rank better wherever both exist, while decision fine-tuning adds little to the untuned model. Decision models are better calibrated only if the question matches the relevance definition, and even then reranker fine-tunes select better passages at a fixed threshold. Our own reranker fine-tune is as well calibrated as the best decision model of its size.

Keywords: Reranking · Calibration · System One Models · Fine-Tuning

1 Introduction

System One models, also called decision models, answer a question about an input (the *state*) with a probability instead of generated text, like fast, intuitive thinking [14]. TypeSafe AI released the first of them, Jev, on 15 September 2026, trained with a new method it calls RLCD (reinforcement learning for calibrated decisions) [31]. The model class is only about a month old, yet it is marketed widely, and first reports show Jev reranking BM25 candidates about as well as strong cross-encoders [11, 13, 15, 26]. Open alternatives with the same kind of output followed within weeks: Kev by the end of September [22] and Cloudflare's Clef on 1 October [4]. They fine-tune open Qwen models, but not with RLCD, which is unpublished [21]. Still, the largest Kev matches Jev as a reranker in our experiments. We therefore study *decision fine-tuning*, i.e. fine-tuning an open model to give Jev's kind of output, and ask, one month after Jev's release, whether it is of use for retrieval.

A fair test compares decision fine-tuning with the established alternative, reranker fine-tuning, on the same *foundation model*, i.e. the same pretrained language model of a given size, such as Qwen3.5-4B. For six Qwen foundation models from 0.6B to 27B parameters and ten test collections, we compare three variants: the base model asked zero-shot, its decision fine-tune and its reranker fine-tune. Since decision models are advertised for their calibrated probabilities, we also check whether they at least deliver that:

RQ1 On the same foundation model, does decision fine-tuning make a better reranker than reranker fine-tuning or no fine-tuning?

RQ2 Are decision models at least better calibrated than reranker fine-tunes of the same foundation model?

Both answers are mostly negative. (1) Wherever both exist (0.6B to 9B), the reranker fine-tune ranks better than the decision fine-tune, also on collections without training data, and up to 9B decision fine-tuning hardly improves on the base model. (2) Decision models are better calibrated than reranker fine-tunes of the same model only if the question matches the relevance definition, and even then they select worse passages at a fixed threshold. Our own reranker fine-tune of Qwen3.5-9B is as well calibrated as the best decision model of its size and ranks as well as the best published reranker. Code, raw outputs and the results on all collections are available.[1]

2 Related Work

Fine-tuned rerankers. Cross-encoders fine-tuned on MS MARCO [19, 20] and LLM rerankers built on LLaMA or Qwen [17, 34, 35] read relevance from a relevance token or a scoring head. Listwise rerankers output a ranking [24, 29], and the Set-Encoder lets passages attend to each other [27]. Abdallah et al. [1] compare many of them on our candidates.

System One models in retrieval. Besides Jev [31], Kev [22], Clef [4] and Laya [5] are open System One models. As rerankers they have only been tried in reports that are not peer-reviewed: Jev on BM25 candidates [11, 13, 15, 26, 32] and Laya as a relevance gate [28]. None compares decision fine-tuning with reranker fine-tuning on the same model.

Calibration. Neural networks tend to be overconfident, and temperature scaling is a simple fix after training [12]; neural rankers are no exception [23]. Calibrated probabilities matter wherever a score is read on an absolute scale, e.g. to cut off a ranked list [2].

3 Experimental Setup

Triplets. For each of six foundation models (Qwen3-0.6B and 8B, Qwen3.5-0.8B, 4B and 9B, and Qwen3.8-27B) we compare three variants. *Base*: the released model, asked the decision models' question in its chat format (thinking off) and scored by the probability of "yes" as the first answer token. *Decision fine-tune*: Kev [21, 22] (LoRA on the pretrained checkpoints up to 9B, a full fine-tune at 27B), Clef-Flash (9B) and Clef (27B) [4]. *Reranker fine-tune*: Qwen3-Reranker [35] (on the same pretrained checkpoints as Kev) and Prism [34], scored by the probability of "yes" against "no"; none is public for 27B. For Qwen3.5-9B we add our own (*Ours-9B*), trained without any decision data: LoRA ($r = 32$),

[1] [TODO: anonymised link]

Table 1. Base model, decision fine-tune and reranker fine-tune of the same foundation model (nDCG@10, ×100, BM25 top-100). Avg: mean over all ten collections, including DL20 and DL22, which are not shown for space; ZS: mean over the collections without a training split (TREC-COVID, Touché, DBPedia). Per foundation model and column: best in bold, second best underlined. Training: ○ base model, ▲/△ decision fine-tune (joint/isolated), ■ reranker fine-tune (rows shaded grey, red, blue); *Ours-9B*: our reranker fine-tune. Paired t-test, Holm-corrected over the ten collections, $p < 0.05$: * reranker fine-tune vs. the decision fine-tune with the best Avg, † that decision fine-tune vs. the base model, ‡ joint vs. isolated. No reranker fine-tune of the 27B model exists.

			TREC DL			BEIR					Mean	
	System	Mode	DL19	DL21	DL23	COVID	NFC	SciF	Touché	DBP	Avg	ZS
Qwen3-0.6B												
○	Qwen3-0.6B		32.6	32.5	14.3	50.3	16.2	18.8	11.5	14.3	23.8	25.4
▲	Kev-0.6B	joint	18.4‡	19.1‡	11.0‡	34.0‡	12.0‡	1.1‡	4.5‡	6.1‡	13.2	14.9
△	Kev-0.6B	isolated	33.4	39.8	18.8	58.2	17.0	16.2	13.0	21.5†	27.0	30.9
■	Qwen3-Reranker-0.6B		**69.5***	**65.2***	**38.8***	**86.2***	**36.4***	**76.6***	**35.7***	**42.7***	**56.1**	**54.9**
Qwen3-8B												
○	Qwen3-8B		70.4	65.8	42.1	80.2	34.1	72.8	24.0	37.1	53.3	47.1
▲	Kev-8B	joint	63.4‡	60.1	35.7	48.2‡	19.8‡	19.2‡	9.7‡	36.7‡	38.8	31.5
△	Kev-8B	isolated	71.2	61.4†	37.6†	80.8	35.2	73.9	26.5	40.0†	52.7	49.1
■	Qwen3-Reranker-8B		**73.2**	**67.0***	**45.2***	**84.0**	**37.3***	**79.9***	**43.1***	**46.2***	**59.4**	**57.8**
Qwen3.5-0.8B												
○	Qwen3.5-0.8B		47.7	53.3	27.9	65.3	20.4	41.1	21.9	25.8	37.6	37.7
▲	Kev-0.8B	joint	35.7	38.6‡	18.2‡	45.3‡	18.6	20.2‡	9.4‡	14.6‡	25.3	23.1
△	Kev-0.8B	isolated	42.8	48.3	23.8	71.9	20.9	36.0	18.0	32.2†	36.0	40.7
■	Prism-0.8B		**72.3***	**66.3***	**43.7***	**84.2***	**36.1***	**76.3***	**35.6***	**41.7***	**57.1**	**53.8**
Qwen3.5-4B												
○	Qwen3.5-4B		69.0	66.2	40.1	79.5	34.1	72.9	24.8	37.0	52.7	47.1
▲	Kev-4B	joint	65.5	62.4	38.3	75.3‡	29.3‡	62.4‡	23.7	40.7	49.8	46.6
△	Kev-4B	isolated	62.6†	62.0	35.4	82.6	34.1	70.7	23.0	39.7†	51.3	48.4
■	Prism-4B		**74.4***	**70.3***	**46.2***	**84.6**	**37.4***	**76.9***	**34.3***	**43.7***	**58.7**	**54.2**
Qwen3.5-9B												
○	Qwen3.5-9B		69.6	67.7	38.1	81.2	33.6	75.0	27.5	35.8	53.7	48.2
▲	Kev-9B	joint	63.1	63.6	39.1	78.2	31.5	60.9‡	22.6	38.5‡	50.2	46.4
△	Kev-9B	isolated	65.7	60.8	35.4	77.7	32.8	72.6	19.8	34.1	50.1	43.8
▲	Clef-Flash	joint	68.8	64.3	40.4	82.1	35.1‡	65.4	21.3	38.8‡	52.8	47.4
△	Clef-Flash	isolated	68.8	66.4	43.8†	82.3	**37.3**†	68.7†	22.1	42.6†	54.7	49.0
■	Prism-9B		**74.0***	**71.0***	44.2	83.0	37.2	**79.8***	34.7*	43.2	59.0	53.7
■	*Ours-9B*		73.8*	67.3	**47.0**	**84.2**	37.1	77.9*	**36.5***	**47.7***	**59.4**	**56.1**
Qwen3.8-27B												
○	Qwen3.8-27B		71.6	68.5	44.1	83.0	37.6	78.5	30.2	45.0	57.4	52.7
▲	Kev-27B	joint	**73.9**	69.9	**49.3**†	**85.9**†	**38.7**†	**81.1**†	**40.8**‡	46.9†	**61.1**	**57.9**
△	Kev-27B	isolated	73.4	**70.1**	46.8	85.5	38.3	79.9	34.0	**47.5**	59.8	55.7
▲	Clef	joint	71.4	68.6	45.9	82.9	36.3	62.9‡	30.8	42.8	56.1	52.2
△	Clef	isolated	69.4	66.7	42.9	83.5	37.8	71.9	33.9	44.1	56.4	53.8

one epoch over 100k MS MARCO training queries with one relevant and 15 hard negative passages each, a ranking loss (InfoNCE) plus a yes/no classification loss (binary cross-entropy). We removed training queries that match a test or development topic and chose the checkpoint on 200 MS MARCO dev queries. Qwen3-Reranker, Prism and Ours-9B saw MS MARCO; Kev's training data contain neither MS MARCO nor BEIR [21].

Collections and candidates. TREC DL 2019–2023 [6–10] (MS MARCO v1/v2 passages [18]) and TREC-COVID, NFCorpus, SciFact, Touché and DBPedia from BEIR [30]. TREC-COVID, Touché and DBPedia have no training split in any fine-tune's data and form the *zero-shot subset* (ZS). Every system reranks the BM25 top-100 of Pyserini [16], the candidates of S1Rank [26], RankGPT [29], RankZephyr [24] and Abdallah et al. [1], whose published numbers we use where we did not rerun a system.

Decision models. All decision models get S1Rank's question [26], fixed in advance ("Is document `{doc}` relevant to the `query`? Relevant means it answers the query or contains information that directly addresses it."), with a state that holds all 100 candidates (*joint*) or one (*isolated*). Besides the triplets we run Jev 1.13.0 (closed), Laya [5] (isolated only, as its context is too short), monoT5-3B [20], bge-reranker-v2-m3 [3], the Set-Encoder [27] and RankZephyr (DL only). For RQ2 we also ask a stricter question on DL 2019–2023, "Does document `{doc}` answer the `query`?", which matches the relevance grade we score; it was fixed before our runs but not chosen on development data.

Metrics. nDCG@10, with paired t-tests over queries, Holm-corrected across the ten collections ($p < 0.05$). Expected calibration error (ECE, 15 bins) [12] on the judged pairs of DL 2019–2023 (relevant from grade 2), the only collections that also judge non-relevant passages. *One cutoff*: we keep every candidate with $p \geq 0.5$ and report the F_1 of this set (unjudged passages count as non-relevant). *Sanity checks*: our BM25 matches Pyserini's reproductions [25], monoT5-3B and RankZephyr are within 0.2 points of their published DL 2019/2020 numbers [1], and Jev joint is within 0.5 of S1Rank's [26] on four of five shared collections (1.5 below on TREC-COVID).

4 Decision vs. Reranker Fine-Tuning

Reranker fine-tuning wins on every model. For all five foundation models where both exist, the reranker fine-tune ranks better than the decision fine-tune in either mode (Table 1; e.g. 58.7 vs. 51.3 for Qwen3.5-4B, and 59.0 for Prism-9B and 59.4 for Ours-9B vs. 54.7 for Clef-Flash). The difference is significant on 46 of the 60 pairs of model and collection, and the decision fine-tune is never significantly better (*). The zero-shot subset shows the same, so the gap does not come from training on MS MARCO, and Ours-9B, trained on MS MARCO alone, matches Prism-9B. Up to 9B, decision fine-tuning hardly changes the base model (between 3.5 points below and 3.2 points above its average; †). Reranker fine-tuning adds 5 to 32 points. Only Kev-27B clearly improves on its base model (by 3.7 points, significant on eight collections). It is the best system in our study (61.1), but only slightly ahead of reranker fine-tunes a third of its size (59.4).

Joint vs. isolated. Scoring all candidates together helps only models trained on long states, and only a little: Kev-27B gains 1.3 points and Jev 1.0 (Table 2),

Table 2. Other systems, same candidates (nDCG@10 ×100, means). †: includes published numbers [1,33]; –: not all collections available. Best in bold, second best underlined; symbols and shading as in Table 1.

	System	Type	DL	BEIR	Avg	ZS
	BM25		39.3	47.1	43.2	45.2
■	RankGPT-4	listwise	–	56.9†	–	<u>57.1</u>†
■	RankZephyr-7B	listwise	61.4	53.3†	57.3†	51.7†
■	monoT5-3B	cross-encoder	60.4	54.6†	57.5†	52.5†
■	bge-reranker-v2-m3	cross-encoder	60.4†	52.8†	56.6†	52.2†
■	Set-Encoder-large	cross-encoder	**63.6**†	54.4	59.0†	53.8
▲	Jev	System One, joint	<u>62.7</u>	**58.2**	**60.5**	**57.2**
△	Jev	System One, isolated	61.4	<u>57.6</u>	<u>59.5</u>	56.7
△	Laya	System One, isolated	47.3	40.7	44.0	43.7

while Kev-0.6B, Kev-0.8B and Kev-8B lose 10.7 to 13.9 points (‡). Kev-0.8B and Kev-4B were trained on states of at most 7,552 tokens and Kev-27B on up to 32,768 [21] (Kev-8B's card gives no length), while a joint state has about 13k tokens on TREC DL and 40k or more on BEIR. Beyond that length the model loses track of which passage a question is about (Kev-0.6B joint is at chance level on DL 2019/2020).

Other systems. The open Kev-27B (61.1) matches the closed Jev (60.5, Table 2), and both lead the established rerankers. Laya (44.0) is barely above BM25 (43.2).

5 Are Decision Models Better Calibrated?

Calibration is the main selling point of decision models. Besides the ECE we report the Brier score, the mean squared error of the probabilities, which also rewards separating relevant from non-relevant passages (Table 3). With S1Rank's question, decision fine-tuning improves on the poorly calibrated base models, but no more than reranker fine-tuning: up to 8B, the reranker fine-tune has a similar or lower ECE and a lower Brier score (e.g. Prism-4B 0.218 and 0.172 vs. Kev-4B 0.216 and 0.207). At 9B, Clef-Flash beats Prism (0.140 vs. 0.242), but Ours-9B is better still (0.111).

The question matters. S1Rank's question also accepts passages that only partly address the query, but we count a passage as relevant only if it answers the query. Asking exactly that (*Answers?*) improves every decision and base model, with hardly any change in ranking. Now the decision fine-tunes beat the published reranker fine-tunes from 4B on (e.g. Kev-8B 0.113 vs. Qwen3-Reranker-8B 0.257). But this is not special to decision fine-tuning: Ours-9B matches Clef-Flash (0.111 vs. 0.112), and the untuned models gain too (Qwen3-8B: 0.383 to 0.231).

Uses of calibration. Better calibration does not make decision models select better passages. If every passage with $p \geq 0.5$ is kept, e.g. for a generator, the

Table 3. Calibration on the judged pairs of TREC DL 2019–2023 (means; lower is better): ECE (15 bins) and Brier score. Decision fine-tunes in their better mode (j/i: joint/isolated); best per foundation model in bold, second underlined. *Relevant?*: S1Rank's question; *Answers?*: whether the passage answers the query; rerankers ask no question (ECE in brackets). Symbols as in Table 1.

System	Relevant? ECE	Relevant? Brier	Answers? ECE
Qwen3-0.6B			
○ Qwen3-0.6B	0.750	0.750	0.750
△ Kev-0.6B (i)	<u>0.579</u>	<u>0.543</u>	0.503
■ Qwen3-Reranker-0.6B	**0.511**	**0.481**	(0.511)
Qwen3-8B			
○ Qwen3-8B	0.383	0.380	0.231
▲ Kev-8B (j)	**0.243**	<u>0.248</u>	0.113
■ Qwen3-Reranker-8B	<u>0.257</u>	**0.238**	(0.257)
Qwen3.5-0.8B			
○ Qwen3.5-0.8B	0.716	0.694	0.617
△ Kev-0.8B (i)	<u>0.357</u>	<u>0.309</u>	0.336
■ Prism-0.8B	**0.243**	**0.193**	(0.243)
Other systems			
▲ Jev (j)	**0.224**	**0.187**	0.075

System	Relevant? ECE	Relevant? Brier	Answers? ECE
Qwen3.5-4B			
○ Qwen3.5-4B	0.296	0.269	0.181
△ Kev-4B (i)	**0.216**	<u>0.207</u>	0.132
■ Prism-4B	<u>0.218</u>	**0.172**	(0.218)
Qwen3.5-9B			
○ Qwen3.5-9B	0.283	0.265	0.165
▲ Kev-9B (j)	0.237	0.209	0.160
▲ Clef-Flash (j)	<u>0.140</u>	<u>0.165</u>	0.112
■ Prism-9B	0.242	0.183	(0.242)
■ *Ours-9B*	**0.111**	**0.157**	(0.111)
Qwen3.8-27B			
○ Qwen3.8-27B	0.166	0.180	0.117
▲ Kev-27B (j)	<u>0.156</u>	<u>0.159</u>	0.076
▲ Clef (j)	**0.147**	**0.157**	0.091

reranker fine-tune keeps the better set on every foundation model, with either question, because it ranks better. Using the probabilities to count a query's relevant passages works best for the untuned Qwen3.8-27B (details in our repository).

6 Conclusion

Do System One models make better rerankers? No. On the same foundation model, reranker fine-tuning beats decision fine-tuning, which adds little to the untuned model. Their calibration is real only with a question that matches the relevance definition, does not make them select better passages, and can be matched by a reranker fine-tune. Our second question was not chosen on development data, and calibration was only measured on TREC DL.

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